```
      }
    }
  }
}
```

We create an index with a custom `english` analyzer and a set of user-defined stop words. When we test a piece of text with this analyzer, the stop words are honored.

Listing 7.20 Testing the custom stop words for the `english` analyzer

```
POST index_with_custom_english_analyzer/_analyze
{
  "text":"A dog is for a life",
  "analyzer":"index_with_custom_english_analyzer"
}
```

This code outputs just two tokens: "dog", and "life". The words *a*, *is*, and *for* were removed because they matched the stop words we specified.

Language analyzers have another feature that they are always eager to implement: stemming. *Stemming* is a mechanism to reduce words to their root form. For example, any form of the word *author* (*authors*, *authoring*, *authored*, etc.) is reduced to the single word *author*. The following listing shows this behavior.

Listing 7.21 Reducing all forms of `author` to the `author` keyword

```
POST index_with_custom_english_analyzer/_analyze
{
  "text":"author authors authoring authored",
  "analyzer":"english"
}
```

This code produces four tokens (based on the `whitespace` tokenizer), all of which are "author"—because the root word for any form of *author* is *author*!

But sometimes, stemming goes too far. If you add "authorization" or "authority" to the list of words in the previous listing, it is also stemmed and indexed as "author"! Obviously, we can't find pertinent answers when searching for "authority" or "authorization" because those words do not make it to the inverted index due to stemming.

All is not lost. We can configure our `english` analyzer, asking it to ignore certain words (such as *authorization* and *authority*, in this case) that do not need to get through the analyzer. We can use the `stem_exclusion` attribute to configure words to exclude from stemming. The next listing does this by creating an index with custom settings and passing the arguments to the `stem_exclusion` parameter.

Listing 7.22 Creating an index with custom stem-exclusion words

```
PUT index_with_stem_exclusion_english_analyzer
{
  "settings": {
```